

Felix D. Priestley

School of Physics and Astronomy, Cardiff University
Queen's Buildings, The Parade, Cardiff CF24 3AA, UK
+44 (0)789 4413 787
priestleyf@cardiff.ac.uk

RESEARCH INTERESTS

- Chemical evolution in star-forming environments
- The formation and destruction of interstellar dust

EMPLOYMENT

Cardiff University, UK **Jul 2019 -**
Post-doctoral research associate

University College London, UK **Oct 2018 - Jun 2019**
Post-doctoral research associate

EDUCATION

University College London, UK **Oct 2015 - Sep 2018**
PhD Astrophysics (awarded Dec 2018)
Thesis Title: Molecule and dust emission at the beginnings and ends of stellar evolution
Supervisor: Prof. Michael Barlow

University College London, UK **Sep 2011 - Jun 2015**
MSci Astrophysics (First class honours)
Dissertation Title: The effects of gravitational collapse on the chemical evolution of prestellar cores
Supervisor: Prof. Serena Viti

PUBLICATION SUMMARY

Refereed papers: 26 first-author, 60 in total
Total citations: 1112, of which 375 from first-author papers

TECHNICAL SKILLS

Programming: Fortran 77/90/Modern (experienced), Python, Linux shell, C/C++
Software: MHD (PHANTOM, AREPO), chemistry (UCLCHEM, UCLPDR), radiative transfer (LIME, RADMC3D), dust emission (DINAMO)

AWARDS

STFC Astronomy Grants Panel Small Award **2025**
Named PDRA on ~£500k grant; contributed majority of research case to proposal

Jon Darius Memorial Prize **2019**
For outstanding postgraduate research in astrophysics at University College London

TALKS & SEMINARS (* - INVITED)

***Leiden Observatory** **May 2026**
Chemical clocks in turbulent molecular clouds

***University of Cologne** **May 2026**
Astrochemistry as a tool for understanding star formation

***UCLCHEM workshop** **Mar 2026**
Collapse and post-processing modules in UCLCHEM

Stellar Origins 2025 <i>Molecular line tracers of dense gas, ‘dense’ gas, and the star formation rate</i>	Sep 2025
*EAS Annual Meeting 2025 <i>The impact of dynamics on the chemistry of star formation</i>	Jun 2025
*Gas in Galaxies workshop <i>Dense gas and the star formation rate on molecular cloud scales</i>	Jun 2025
*Oxford University <i>Connecting molecular line emission with the star formation rate</i>	Oct 2024
EAS Annual Meeting 2024 <i>Understanding complex organic molecules in the earliest phases of star formation</i>	Jul 2024
Early Phases of Star Formation 2024 <i>Molecular tracers of the threshold density for star formation</i>	May 2024
Centre for Astrochemical Studies, MPE <i>Modelling chemical evolution in molecular clouds self-consistently</i>	Jan 2024
*ECOGAL collaboration seminar series <i>Chemical evolution in molecular clouds</i>	Nov 2023
Leiden Observatory <i>Modelling chemical evolution in molecular clouds self-consistently</i>	Nov 2023
Origin and Fate of Dust in Our Universe <i>Empirical constraints on dust destruction in supernova remnants</i>	Sep 2023
*AREPO ISM development workshop <i>Post-processing chemical evolution in hydrodynamical simulations</i>	Sep 2023
National Astronomy Meeting 2023 <i>Unveiling the origins of prestellar cores with molecular line emission</i>	Jul 2023
The Physics of Star Formation <i>Can prestellar cores be modelled as isolated objects?</i>	Jun 2023
*Universidad Complutense Madrid <i>Testing theories of star formation with molecular line data</i>	Mar 2023
University College London <i>Probing the importance of magnetic fields in star-forming regions using molecular line emission</i>	Mar 2022
*St. Andrews University <i>What can molecular lines tell us about star formation?</i>	Jan 2022
National Astronomy Meeting 2021 <i>The properties of shocked dust in supernova remnants</i>	Jul 2021
Magnetic fields and the structure of the filamentary ISM <i>The characteristic widths of magnetised filaments</i>	Jun 2021
ISM Scales 2021 <i>Filament widths in molecular clouds: are they universal, and if so, why?</i>	May 2021

	*Supernovae and Interstellar Dust workshop <i>Observational constraints on dust destruction in shocks</i>	Apr 2021
	The Rise of Metals and Dust in Galaxies through Cosmic Time <i>Cold dust emission from the shocked material around supernova remnants</i>	Oct 2020
	European Week of Astronomy and Space Science 2019 <i>The survival of dust grains in the ejecta of core-collapse supernovae</i>	Jun 2019
	*Cardiff University <i>Molecular tracers of star formation mechanisms</i>	Apr 2019
	The Supernova-Supernova Remnant Connection <i>The pre- and post-shock dust mass in Cassiopeia A</i>	Jan 2019
SUPERVISION	Christopher Le Quesne PhD co-supervisor	2026 -
	Oliver Biggs PhD co-supervisor	2025 -
	Jimitbhai Panchal MSc research project supervisor	2025
	Lillian Cai Undergraduate research internship; submitted to OJAp	2025
	Rees Barnes Undergraduate research internship; published in OJAp	2024
	Charles Yin MSc research project supervisor; published in MNRAS	2020
TEACHING	Project supervisor, Cardiff University Supervision of third-year undergraduate research projects on N-body dynamics	2025 -
	Cardiff Astrophysical Summer School Lecture on Linux systems and introduction to command-line programming	2025
	Deputy module organiser, Cardiff University Administrative and teaching duties for undergraduate maths course (~100 students)	2022 - 2024
	Demonstrator, Cardiff University Senior lab demonstrator for undergraduate observational astronomy course	2021 - 2024
	Demonstrator, University of London Observatory Lab demonstrator for undergraduate practical astronomy courses	2013 - 2017
COMMUNITY	External reviewer for STFC Astronomy Grants Panel	2023 -
	Referee for ApJ, MNRAS, A&A, Nature Astronomy	2020 -
	Seminar organiser, Cardiff Astronomy group	2019 - 2022, 2025 -

	Cardiff Physics & Astronomy research committee member	2025 -
	AREPO code development team	2023 -
	BISTRO collaboration member	2020 -
	Co-developer of UCLCHEM and UCLPDR codes	2017 -
	Focus group organiser, EPoS 2024	2024
	SOC for Cosmic Star Formation session, NAM 2021	2021
OUTREACH	Barry Astronomical Society Public talk: Molecules in space: the occasional relevance of astrochemistry	Apr 2025
	Astronomy on Tap, Cardiff Public talk: A brief history of star formation	Jun 2024
	Howell's School, Cardiff Public talk: Where do stars come from?	Mar 2023
	Barry Astronomical Society Public talk: Cores, clouds and filaments: where do stars form, and why?	Feb 2022
	Royal Society Summer Science Exhibition JWST exhibit demonstrator	Jul 2018
	Cafe Scientifique Public talk: Cosmic Dust from Exploding Stars	May 2017
REFeree CONTACT INFORMATION	Prof. Paul Clark School of Physics and Astronomy, Cardiff University Queens Buildings, The Parade, Cardiff CF24 3AA, UK clarkpc@cardiff.ac.uk	
	Prof. Michael Barlow Department of Physics and Astronomy, University College London Gower Street, London, WC1E 6BT, UK mjb@star.ucl.ac.uk	
	Prof. Serena Viti Leiden Observatory, Leiden University P.O. Box 9513, 2300 RA Leiden, The Netherlands viti@strw.leidenuniv.nl	
	Prof. Ralf Klessen Institut für Theoretische Astrophysik, Universität Heidelberg Albert-Ueberle-Straße 2, D-69120 Heidelberg, Germany klessen@uni-heidelberg.de	
	Prof. Simon Glover Institut für Theoretische Astrophysik, Universität Heidelberg Albert-Ueberle-Straße 2, D-69120 Heidelberg, Germany glover@uni-heidelberg.de	

Prof. Ilse De Looze

Sterrenkundig Observatorium, Ghent University
Krijgslaan 281 - S9, 9000 Gent, Belgium
ilse.delooze@ugent.be

Prof. Anthony Whitworth

School of Physics and Astronomy, Cardiff University
Queens Buildings, The Parade, Cardiff CF24 3AA, UK
anthony.whitworth@astro.cf.ac.uk